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GRADE A+**
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Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 2

Title: Superstore Sales Performance Analysis Dashboard

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Superstore Sales Performance Analysis Dashboard

Problem Statement

Management needs a simple and interactive way to understand the company's sales performance. The objective of this project is to analyze sales data across **products, categories, regions**, customer segments, and time periods.

The Power BI dashboard is developed to help management quickly identify:

- Overall sales and profit performance
- Best-performing product categories
- Top-selling products
- Regional sales performance
- Sales trends over time
- Customer segment performance

The dashboard helps convert raw business data into meaningful visual insights for better decisionmaking

Step 1: Data Collection

The Sample Superstore dataset was collected from Kaggle.

The dataset contains business transaction information such as:

- Order ID
- Order Date
- Customer information
- Product information
- Category
- Sub-Category
- Region
- Sales
- Profit
- Quantity
- Discount
- Ship Mode
- Segment

The dataset was selected because it provides sufficient information to analyze different aspects of business sales performance.

Step 3: Data Import into Power BI

The dataset was imported into Microsoft Power BI Desktop.

Steps followed:

1. Open Power BI Desktop.
2. Select Get Data.

3. Select the appropriate file format.
4. Select the Sample Superstore dataset.
5. Load the data into Power BI.
6. Verify the imported columns and records.

After importing the dataset, the data was available for further cleaning, transformation, analysis, and visualization.

Step 4: Data Cleaning and Transformation

The imported data was checked and prepared using Power Query.

The main data preparation activities included:

- Checking column names
- Checking data types
- Converting date fields into appropriate date formats
- Checking numerical fields such as Sales, Profit and Quantity
- Checking categorical fields such as Category and Region
- Removing unnecessary or duplicate information if required
- Ensuring the dataset was suitable for visualization
- Used the Sales, Product and Customer tables for analysis.
- Connected the tables using key fields such as ProductKey and CustomerKey.
- Used OrderDate for time-based analysis.
- Created a data model to analyze sales based on products, customers and dates.
- Used the model to build the AdventureWorks Sales Dashboard.

Step 5: Data Modeling

The Sample Superstore dataset was organized and prepared for analysis using Power BI. The data contains information related to customers, products, orders, sales, profit, quantity, discount, segment, ship mode, city, state, and region.

The data model was used to analyze business performance from different perspectives.

- Used **Order Date** for date-based analysis.
- Used **Customer ID** for customer-level analysis.
- Used **Product ID and Product Name** for product analysis.
- Used **City and State** for geographical analysis.
- Used **Region** for regional comparison.
- Used **Segment** to compare different customer groups.
- Used **Ship Mode** to analyze shipping performance.
- Used **Sales, Profit, Quantity, and Discount** for business performance analysis.

Step 6: DAX Measures

DAX (Data Analysis Expressions) was used to create the calculations required for the dashboard.

Total Orders

Total Orders = DISTINCTCOUNT('Sample_Superstore'[Order ID])

This measure calculates the total number of unique orders.

The dashboard also uses Power BI aggregation for:

- Total Sales → SUM(Sales)
- Total Profit → SUM(Profit)
- Total Quantity → SUM(Quantity)
- Total Discount → SUM(Discount)

These measures are used in different charts and KPI visuals to understand overall business performance.

Step 7: Dashboard Development

The dashboard was developed in Power BI using multiple visualizations. The final dashboard contains the following components:

Dashboard Visualizations

- Sum of Discount by City
- Sum of Discount, Profit and Sales by Segment, Sub-Category and Ship Mode
- Sum of Discount and First Product ID by Order Date
- Sum of Sales, Quantity, Profit and Discount
- Sum of Discount KPI/Doughnut Visual
- Count of Row ID by Ship Mode and State
- Customer ID, Country and State Table
- Sum of Profit, Quantity and Sales by Region

These visualizations provide a detailed view of sales, profitability, discounts, customer information, shipping methods, and regional performance.

Step 8: Interactive Filters and Slicers

An interactive State, Country slicer was added to the right side of the dashboard.

The slicer allows users to select specific states and analyze the corresponding data.

The dashboard can be used to interactively analyze:

- State
- Country
- City
- Region
- Ship Mode
- Segment
- Sub-Category
- Order Date

When a user selects a particular state or country, the related dashboard visuals are automatically updated.

Step 9: Dashboard Formatting

Formatting was applied to make the dashboard simple, clean, and easy to understand.

The following formatting techniques were used:

- Used a clean and simple dashboard layout.
- Arranged the visualizations in organized sections.
- Added meaningful titles to each visual.
- Used consistent Power BI formatting.
- Added an interactive State and Country slicer.
- Used appropriate chart types for different analyses.
- Displayed important numerical values using KPI-style visuals.
- Maintained consistent fonts and spacing.
- Avoided unnecessary visual elements.

The dashboard was designed so that important business information can be understood quickly.

Step 10: Analysis and Insights

The dashboard provides several important business insights.

Discount Performance

The central discount visual shows a total discount of approximately 1.56K. This provides an overview of the discount amount represented in the dashboard.

City Performance

The Sum of Discount by City chart shows that Philadelphia has the highest visible discount, followed by Houston and Chicago.

Regional Performance

The regional chart compares Sales, Profit and Quantity across the four regions:

- West
- East
- South
- Central

The West and East regions show strong sales performance compared with the other displayed regions.

Ship Mode and State

The Count of Row ID by Ship Mode and State visualization compares order volume across different shipping methods and states.

The chart shows that Standard Class has the highest order count among the displayed ship modes.

Customer Details

The customer table displays:

- Customer ID
- Country
- State

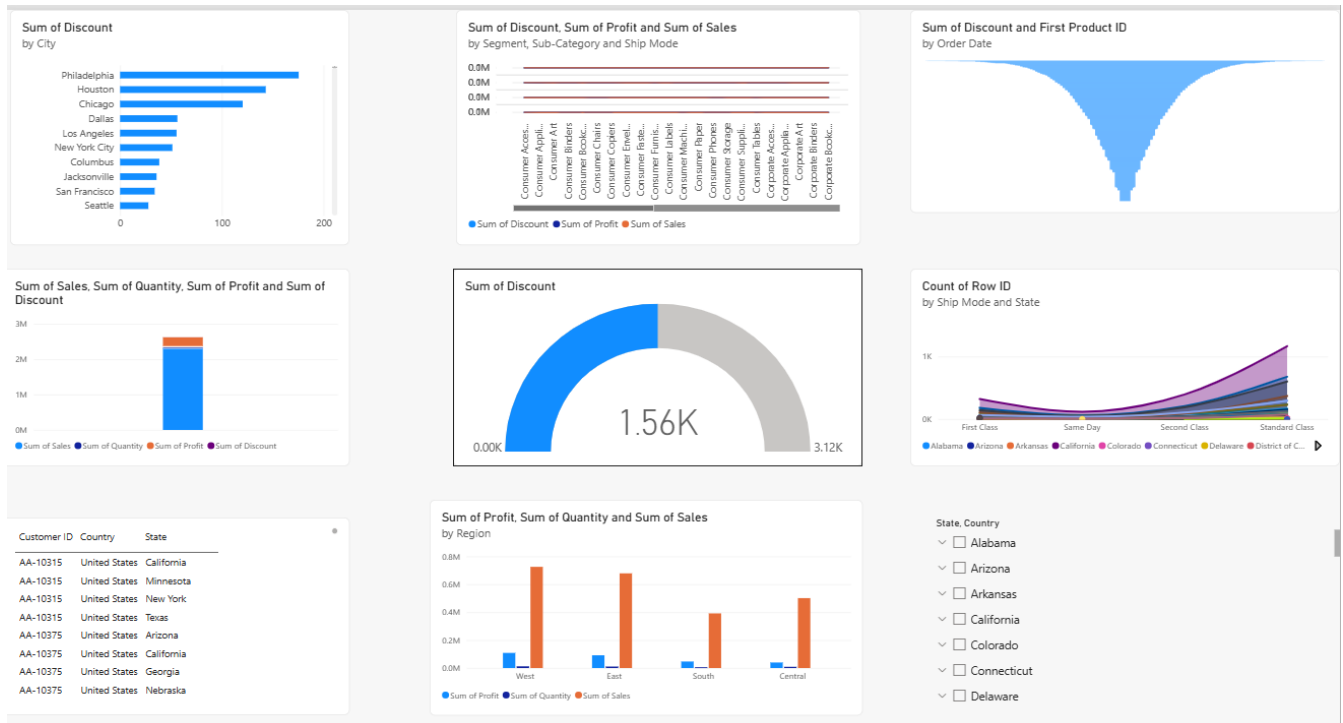
This allows users to examine customer records according to their geographical location.

Time-Based Analysis

The Order Date visualization provides date-based analysis of discount and product information, helping users observe changes across different dates.

Segment and Product Analysis

The dashboard combines Segment, Sub-Category and Ship Mode with Sales, Profit and Discount to provide detailed performance comparisons across different business dimensions.



Step 11: Final Dashboard

The final dashboard combines important business information into a single interactive Power BI report.

Main Components

Discount Analysis

- Sum of Discount by City
- Sum of Discount by Order Date
- Overall Discount KPI

Sales and Profit Analysis

- Sales
- Profit
- Quantity
- Discount
- Regional comparison

Product and Customer Analysis

- Product ID
- Product-related analysis
- Customer ID
- Country
- State

Shipping Analysis

- Ship Mode
- State-wise order count

Regional Analysis

- Sales by Region
- Profit by Region
- Quantity by Region

Interactive Controls

- State
- Country

The final dashboard allows management to interact with the data and quickly identify important patterns in sales, profit, discount, quantity, customers, products, regions, states, and shipping methods.

Learning Outcomes

Through this project, the following skills were developed:

- Understanding business datasets
- Data import using Power BI
- Data cleaning using Power Query
- Data transformation
- Creating DAX measures
- Creating KPI cards
- Creating different Power BI visualizations
- Creating interactive slicers
- Dashboard design and formatting
- Business data analysis
- Identifying patterns and trends
- Presenting data-driven insights
- Using Power BI as a business intelligence tool

